

Text Emotion Classification using Convolutional Neural Networks

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Abstract. This paper presents an experimental comparison of two neural network architectures for emotion classification in text: TextCNN and Bidirectional LSTM (BiLSTM). We use the dair-ai/emotion dataset containing 20,000 English sentences labeled with six emotion categories: sadness, joy, love, anger, fear, and surprise. Both models utilize pre-trained GloVe 300-dimensional word embeddings. We conduct systematic experiments analyzing the impact of convolutional kernel sizes, network depth, and regularization. Our results show that BiLSTM achieves superior performance with F1 macro score of 0.885 compared to TextCNN with 0.853, demonstrating that sequential context modeling is more effective for emotion detection in short texts. Ablation studies confirm that kernel size diversity in TextCNN has minimal impact, while network depth in BiLSTM is critical for performance.

Keywords: emotion classification · TextCNN · BiLSTM · GloVe · natural language processing

1 Problem Formulation

The task of automatic emotion recognition in text is a fundamental problem in natural language processing with applications in sentiment monitoring, mental health analysis, and human-computer interaction. Given a short English sentence, the goal is to classify it into one of six emotion categories: sadness, joy, love, anger, fear, or surprise.

Formally, given input sentence $x = (w_1, w_2, \dots, w_n)$ where w_i are word tokens, the model learns a function $f : x \rightarrow y$ where $y \in \{0, 1, 2, 3, 4, 5\}$ corresponds to one of the six emotion classes.

We compare two fundamentally different neural network approaches: a convolutional approach (TextCNN) that captures local n-gram patterns, and a recurrent approach (BiLSTM) that models sequential dependencies across the entire sentence.

2 Theoretical Background

2.1 Word Embeddings

Both models use pretrained GloVe (Global Vectors for Word Representation) embeddings with 300 dimensions trained on 6 billion tokens from Wikipedia and

Gigaword. GloVe embeddings encode semantic relationships between words as dense vectors, where semantically similar words have similar vector representations. Embeddings are kept frozen during training to prevent overfitting on the relatively small dataset.

2.2 TextCNN

TextCNN applies 1D convolutional filters of varying sizes over word embedding sequences. Each filter of size k operates as a sliding window over k consecutive words, learning to detect local patterns (n-grams) that are indicative of specific emotions. Global Max Pooling then selects the most prominent feature detected by each filter across the entire sentence, regardless of its position. This makes TextCNN inherently position-invariant.

2.3 Bidirectional LSTM

BiLSTM processes the input sequence in both forward and backward directions simultaneously. At each time step t , the LSTM cell updates its hidden state based on the current input and the previous hidden state, selectively remembering or forgetting information through gating mechanisms. The bidirectional variant concatenates hidden states from both directions, capturing both past and future context for each word.

3 Dataset and Preprocessing

3.1 Dataset

We use the dair-ai/emotion dataset, which contains English Twitter messages annotated with six basic emotions. The dataset is pre-split into training (16,000 samples), validation (2,000 samples), and test (2,000 samples) sets.

As shown in Fig. 1, the dataset exhibits significant class imbalance: joy (33.5%) and sadness (29.2%) dominate, while surprise (3.6%) and love (8.2%) are underrepresented. This imbalance is addressed during training using weighted cross-entropy loss.

3.2 Preprocessing

Text preprocessing consists of the following steps:

1. Lowercasing all text
2. Removing special characters, keeping only alphanumeric characters and spaces
3. Tokenization by whitespace
4. Building vocabulary from all splits (17,098 unique tokens)
5. Loading GloVe 300d embeddings with 92.9% vocabulary coverage
6. Padding/truncating sequences to maximum length of 50 tokens

Average sentence length in the dataset is 19.2 words, with 95% of sentences containing fewer than 41 words, making MAX_SEQ_LEN=50 sufficient to cover the vast majority of samples.

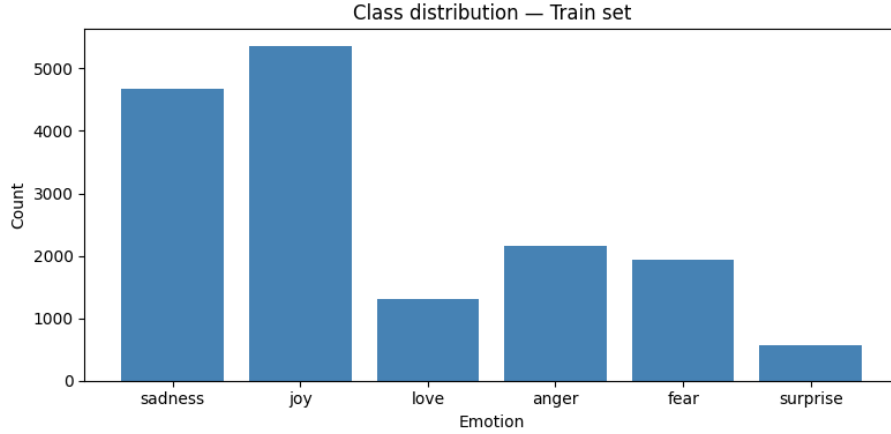


Fig. 1. Class distribution in the training set.

4 Methodology

4.1 TextCNN Architecture

The TextCNN model consists of:

- Embedding layer initialized with GloVe weights (frozen, 300-dim)
- Parallel Conv1D layers with multiple kernel sizes
- ReLU activation after each convolution
- Global Max Pooling over time for each filter
- Concatenation of all pooled features
- Dropout layer for regularization
- Fully connected output layer (6 classes)

4.2 BiLSTM Architecture

The BiLSTM model consists of:

- Embedding layer initialized with GloVe weights (frozen, 300-dim)
- Bidirectional LSTM with configurable hidden size and number of layers
- Concatenation of final hidden states from forward and backward directions
- Dropout layer for regularization
- Fully connected output layer (6 classes)

4.3 Training Setup

Both models share identical training configuration for fair comparison:

- Optimizer: Adam with learning rate 0.001

- Loss: CrossEntropyLoss with class weights (inversely proportional to class frequency)
- Batch size: 64
- Maximum epochs: 20
- Early stopping: patience = 3 epochs based on validation F1 macro
- Evaluation metric: F1 macro (preferred over accuracy due to class imbalance)

5 Experimental Design

5.1 TextCNN Experiments

We conduct three experiments varying kernel sizes and number of filters:

Table 1. TextCNN experiment configurations.

Experiment	Kernel sizes	Filters
Exp1 – Baseline	[3, 4, 5]	100
Exp2 – Large	[2, 3, 4, 5]	200
Exp3 – Ablation (single kernel)	[3]	100

Exp3 serves as an ablation study to isolate the contribution of kernel size diversity.

5.2 BiLSTM Experiments

We conduct three experiments varying hidden size and number of layers:

Table 2. BiLSTM experiment configurations.

Experiment	Hidden size	Layers
Exp1 – Baseline	128	2
Exp2 – Large	256	2
Exp3 – Ablation (single layer)	128	1

6 Results

6.1 TextCNN Results

6.2 BiLSTM Results

7 Analysis and Discussion

7.1 BiLSTM vs TextCNN

BiLSTM consistently outperforms TextCNN across all experiments, achieving F1 macro of 0.885 versus 0.853 for the best TextCNN configuration. This sug-

Table 3. TextCNN test set results.

Experiment	Accuracy	F1 macro	Epochs
Exp1 – Baseline [3,4,5]	88.90%	0.8525	11
Exp2 – Large [2,3,4,5]	88.65%	0.8469	12
Exp3 – Ablation [3]	89.20%	0.8535	19

Table 4. TextCNN Exp1 per-class F1 scores.

Class	F1 score
sadness	0.9215
joy	0.9073
love	0.7886
anger	0.8978
fear	0.8736
surprise	0.7261

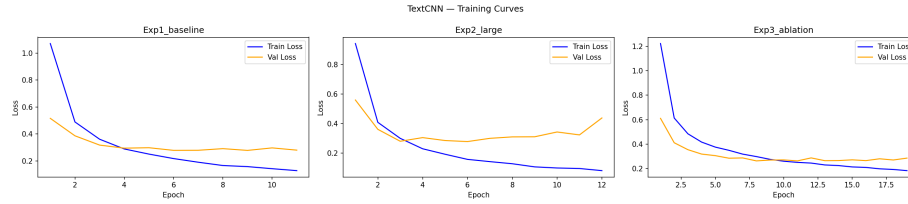
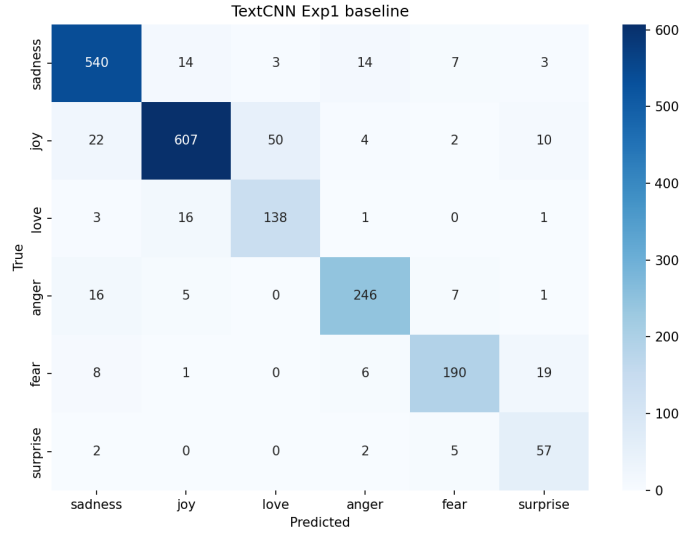
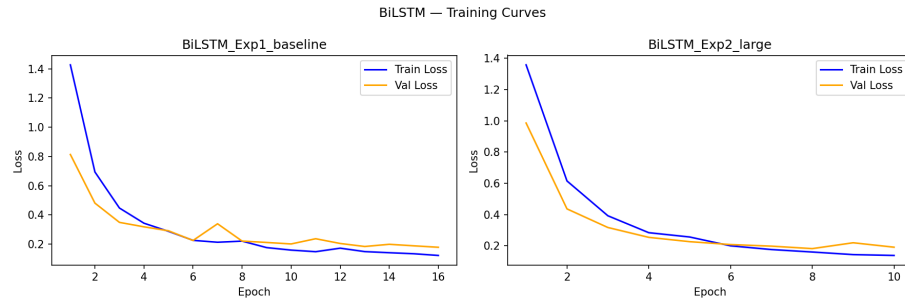
**Fig. 2.** TextCNN training and validation loss curves for all three experiments.**Fig. 3.** TextCNN Exp1 confusion matrix on the test set.

Table 5. BiLSTM test set results.

Experiment	Accuracy	F1 macro	Epochs
Exp1 – Baseline hidden=128	91.55%	0.8851	16
Exp2 – Large hidden=256	91.45%	0.8816	10
Exp3 – Ablation 1 layer	88.20%	0.8491	11

Table 6. BiLSTM Exp1 per-class F1 scores.

Class	F1 score
sadness	0.9532
joy	0.9290
love	0.8278
anger	0.9239
fear	0.8838
surprise	0.7927

**Fig. 4.** BiLSTM training and validation loss curves for all three experiments.

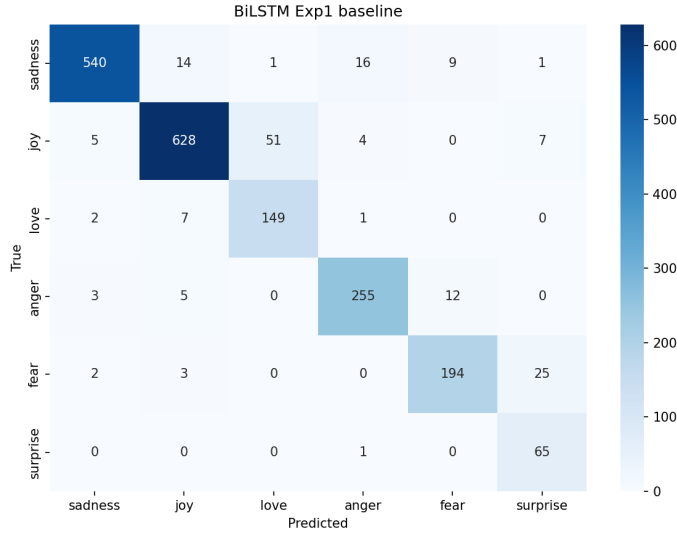


Fig. 5. BiLSTM Exp1 confusion matrix on the test set.

gests that sequential modeling of word dependencies is more effective than local pattern detection for emotion classification in short texts. Emotion expression often relies on sentence-level context — for example, negation ("I do not feel happy") changes the meaning of local patterns that CNN might misclassify.

7.2 Impact of Kernel Size (TextCNN Ablation)

Surprisingly, using a single kernel size [3] achieves comparable performance (F1 0.8535) to multiple kernel sizes [3,4,5] (F1 0.8525). This can be explained by the short average sentence length (19.2 words): with such brief texts, all kernel sizes capture similar contextual patterns. The ablation study suggests that for short text emotion classification, kernel size diversity provides negligible benefit.

7.3 Impact of Network Depth (BiLSTM Ablation)

Reducing BiLSTM from 2 layers to 1 layer causes a significant performance drop from F1 0.885 to 0.849 (−4.2%). This demonstrates that the second LSTM layer learns higher-level representations that are crucial for distinguishing between semantically similar emotions such as fear and sadness.

7.4 Error Analysis

Both models struggle most with minority classes: surprise (F1 0.73 for TextCNN, 0.79 for BiLSTM) and love (F1 0.79, 0.83). This is expected given the severe class

imbalance — surprise represents only 3.6% of training data. Confusion matrix analysis reveals that surprise is most frequently confused with joy, and love with joy, suggesting semantic overlap between positive emotion categories.

8 Limitations and Future Work

The main limitations of this work are:

- Class imbalance remains challenging despite weighted loss — future work could explore oversampling techniques or data augmentation
- GloVe embeddings are frozen — fine-tuning or using contextual embeddings (BERT) could improve performance
- Dataset is limited to Twitter-style short texts — generalization to other domains is uncertain

Future work could explore transformer-based models such as BERT or RoBERTa, which have shown state-of-the-art performance on emotion classification tasks.